第 7 节 实操 任务1：配置 GitHub Actions ...

> **目标**：daily-collect.yml 推送成功 + Actions 手动触发成功

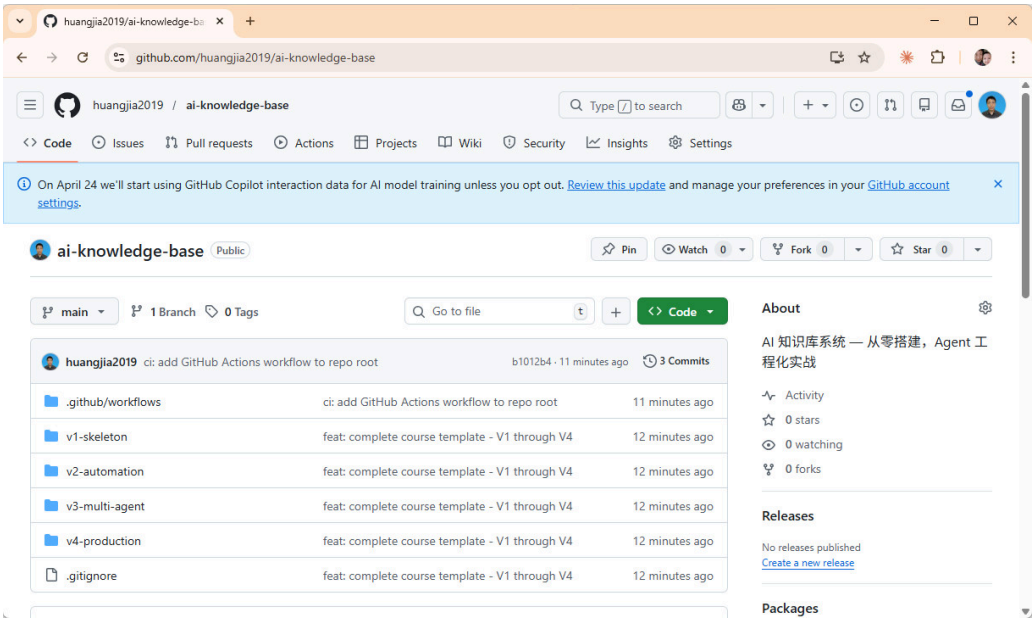
1.1 前置条件

确认你的项目已推送到 GitHub：

```
cd ~/ai-knowledge-base
git remote -v
# 应该看到 origin 指向你的 GitHub 仓库
```

如果还没有推送：

```
# 在 GitHub 上创建仓库（不勾选 Initialize with README）
git remote add origin https://github.com/你的用户名/ai-knowledge-base.git
git push -u origin main
```



1.2 用 AI 编程工具生成 workflow 文件

> 以下配置可以用 **OpenCode**、**Claude Code**、**Cursor**、**Trae** 或**通义灵码**等任意 AI 编程工具生成。

提示词：

```
请帮我创建 .github/workflows/daily-collect.yml，一个 GitHub Actions 工作流：

需求：
```

```
1. 每天 UTC 08:00（北京时间 16:00）自动运行
2. 同时支持手动触发（workflow_dispatch）
3. 添加 permissions: contents: write
4. 使用 Python 3.11，启用 pip 缓存
5. 通过 pip install -r requirements.txt 安装依赖
6. 运行命令：python pipeline/pipeline.py --sources github,rss --limit 20 -
7. 支持多个 LLM 密钥（LLM_PROVIDER、DEEPSEEK_API_KEY、QWEN_API_KEY、OPENAI_
8. 采集后运行 validate_json.py 和 check_quality.py 校验文章
9. 自动 git commit + push，commit 消息包含文章数量和日期
10. 如果没有新数据则不提交（避免空 commit）
```

生成的配置：（参考实现）

```yaml
name: Daily AI Knowledge Collection

# 每天 UTC 08:00（北京时间 16:00）自动采集
on:
  schedule:
    - cron: "0 8 * * *"
  workflow_dispatch:  # 支持手动触发

permissions:
  contents: write

jobs:
  collect:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout repository
        uses: actions/checkout@v4

      - name: Setup Python
        uses: actions/setup-python@v5
        with:
          python-version: "3.11"
          cache: "pip"

      - name: Install dependencies
        run: pip install -r requirements.txt

      - name: Run collection pipeline
        env:
          LLM_PROVIDER: ${{ secrets.LLM_PROVIDER }}
          DEEPSEEK_API_KEY: ${{ secrets.DEEPSEEK_API_KEY }}
          QWEN_API_KEY: ${{ secrets.QWEN_API_KEY }}
          OPENAI_API_KEY: ${{ secrets.OPENAI_API_KEY }}
          GITHUB_TOKEN: ${{ secrets.GITHUB_TOKEN }}
        run: |
```

```
          python pipeline/pipeline.py \
            --sources github,rss \
            --limit 20 \
            --verbose

      - name: Validate collected articles
        run: |
          if ls knowledge/articles/*.json 1>/dev/null 2>&1; then
            python hooks/validate_json.py knowledge/articles/*.json
            python hooks/check_quality.py knowledge/articles/*.json
          else
            echo "No articles to validate"
          fi

      - name: Commit and push results
        run: |
          git config user.name "github-actions[bot]"
          git config user.email "github-actions[bot]@users.noreply.githu

          git add knowledge/
          if git diff --staged --quiet; then
            echo "No new articles to commit"
          else
            ARTICLE_COUNT=$(git diff --staged --name-only | grep -c '\.j
            git commit -m "chore: daily collection - ${ARTICLE_COUNT} ar
            git push
          fi
```

> 如果你对这段配置有疑问，可以让 AI 编程工具解释：
>
> `请解释 daily-collect.yml 中每一步的作用：`
>
> `1. cron: "0 8 * * *" 是什么意思？为什么选 UTC 08:00？`
>
> `2. permissions: contents: write 为什么需要？`
>
> `3. 为什么用 pip install -r requirements.txt 而不是单独装包？`
>
> `4. Validate 步骤的 if ls ... 判断有什么作用？`
>
> `5. commit 消息里的 ARTICLE_COUNT 是怎么算出来的？`

逐行理解：

行	含义
cron: "0 8 * * *"	每天 UTC 08:00 触发（北京时间 16:00）
workflow_dispatch	允许从 GitHub 页面手动触发
permissions: contents: write	授权 workflow 推送代码
actions/checkout@v4	拉取仓库代码到 runner
actions/setup-python@v5	安装 Python 3.11，启用 pip 缓存
pip install -r requirements.txt	安装项目所有依赖
--sources github,rss --limit 20	从 GitHub 和 RSS 源采集，每次最多 20 条
validate_json.py + check_quality.py	采集后自动校验文章格式和质量
ARTICLE_COUNT	统计本次新增的 JSON 文件数量
git diff --staged --quiet	如果有新数据才 commit（避免空提交）

1.3 创建目录并保存文件

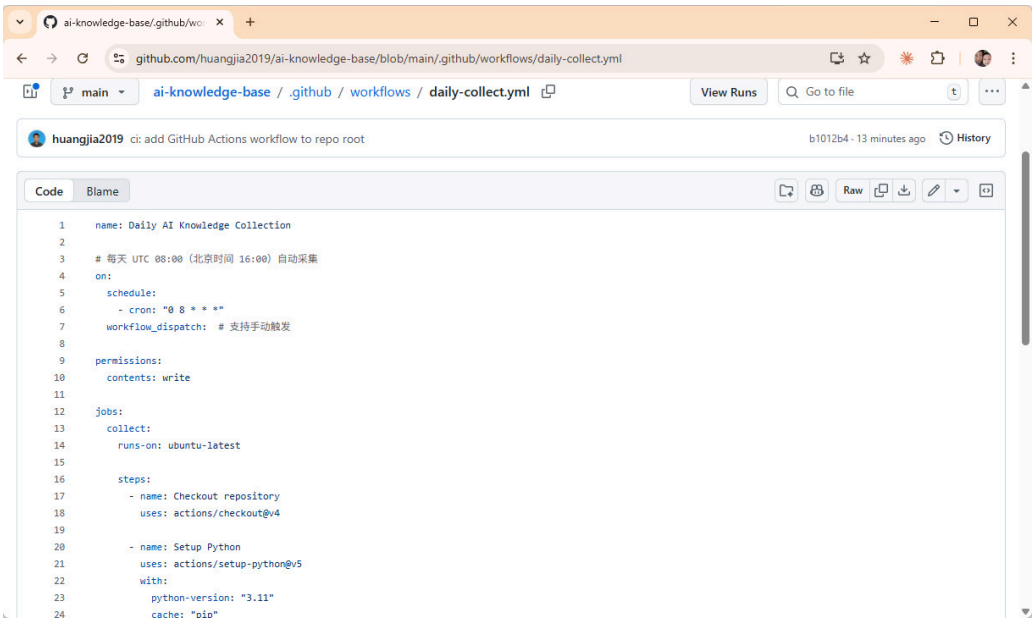
```
mkdir -p .github/workflows
```

将上面的 YAML 内容保存到 `.github/workflows/daily-collect.yml`。

1.4 推送到 GitHub

```
git add .github/workflows/daily-collect.yml
git commit -m "ci: add daily collection workflow"
git push
```

> **注意**：因为我的Repo里面有多个目录，而工作流只能在根目录下面跑，因此所有路径都要加 `v2-automation/` 前缀。这样所有命令都在 v2 子目录下跑，最后 commit 步骤切回根目录 push。



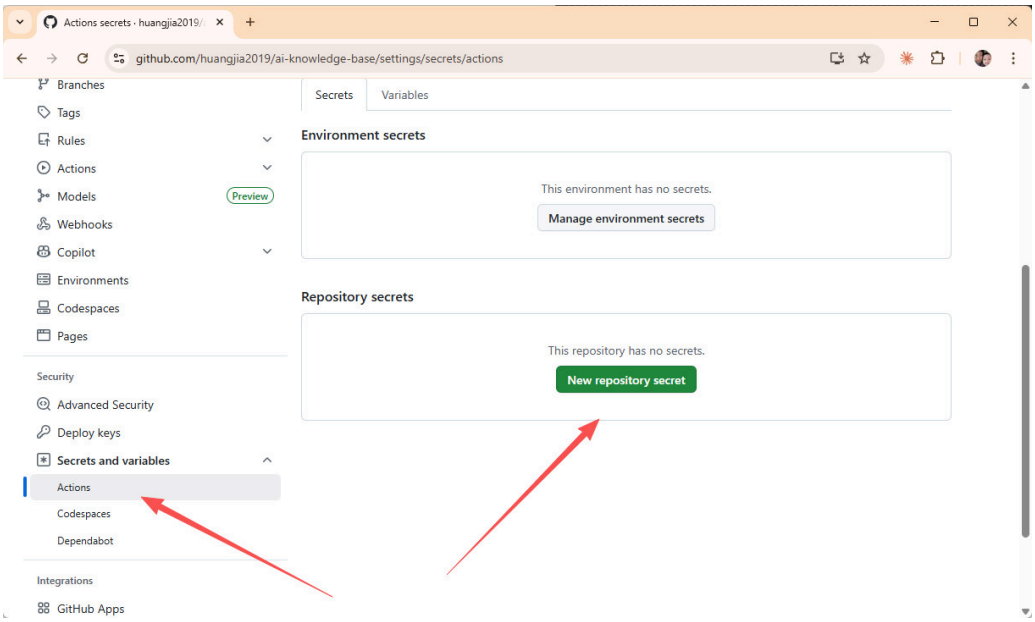
1.5 配置 GitHub Secrets

Workflow 运行时需要 API Key，但**绝对不能写在代码里**。GitHub Secrets 是专门管密钥的地方。

1. 打开 GitHub 仓库页面
2. 点击顶部 **Settings** 标签
3. 左侧菜单：**Secrets and variables → Actions**
4. 点击 **New repository secret**，逐个添加：

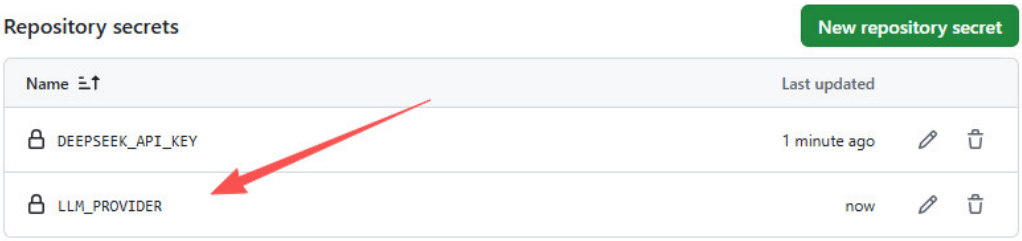
Secret 名称	值	必填
`DEEPSEEK_API_KEY`	你的 DeepSeek API Key（`sk-...`）	是
`LLM_PROVIDER`	`deepseek`	是
`QWEN_API_KEY`	你的 Qwen API Key（如果用 Qwen）	否

5. 每个 Secret 填完点 **Add secret** 保存



检查清单：

检查项	期望	实际
DEEPSEEK_API_KEY 已添加	是	
LLM_PROVIDER 已添加	是	
Secret 值不含多余空格	是	

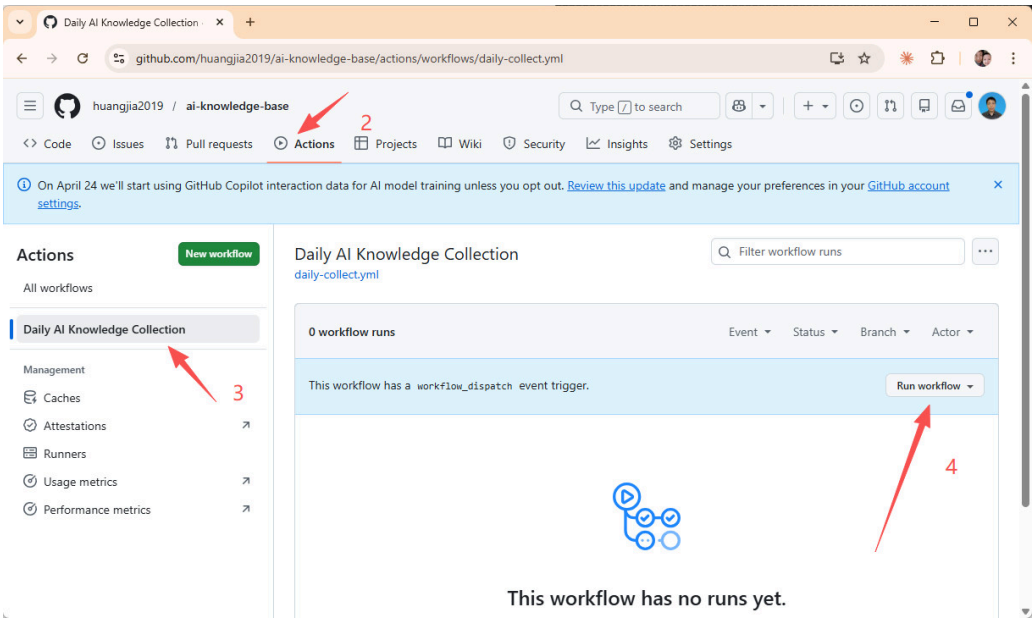


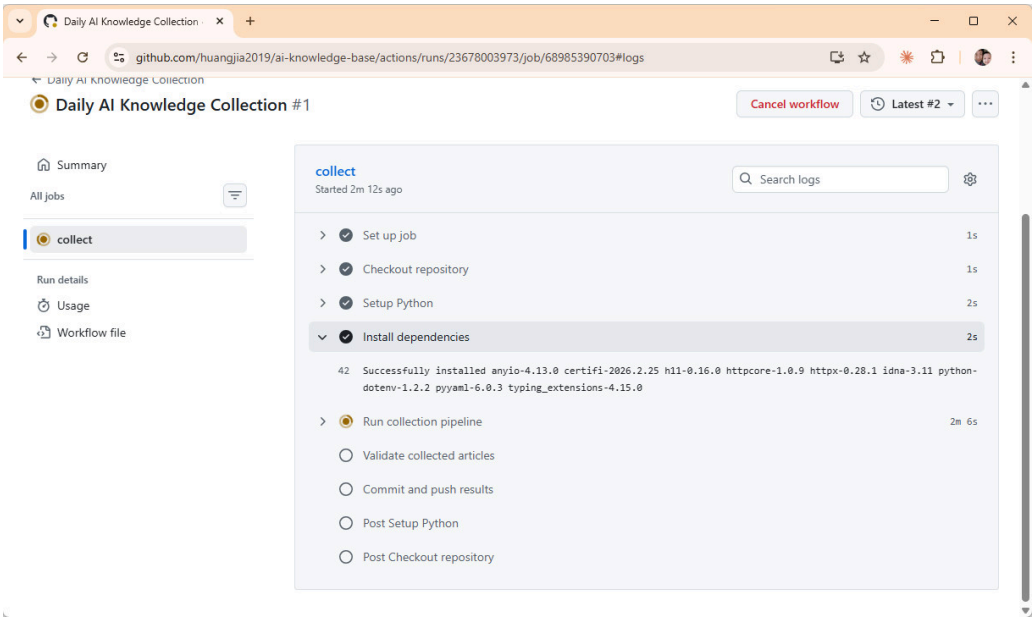
1.6 手动触发测试

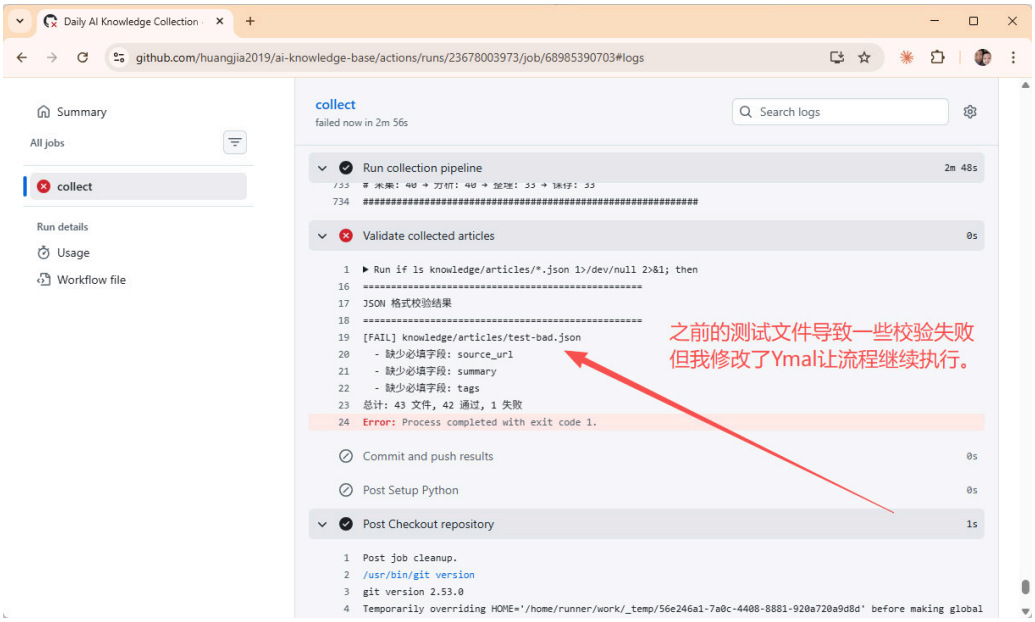
1. 打开 GitHub 仓库页面
2. 点击顶部 **Actions** 标签
3. 左侧选择 **Daily Knowledge Collection**
4. 点击 **Run workflow** -> **Run workflow**

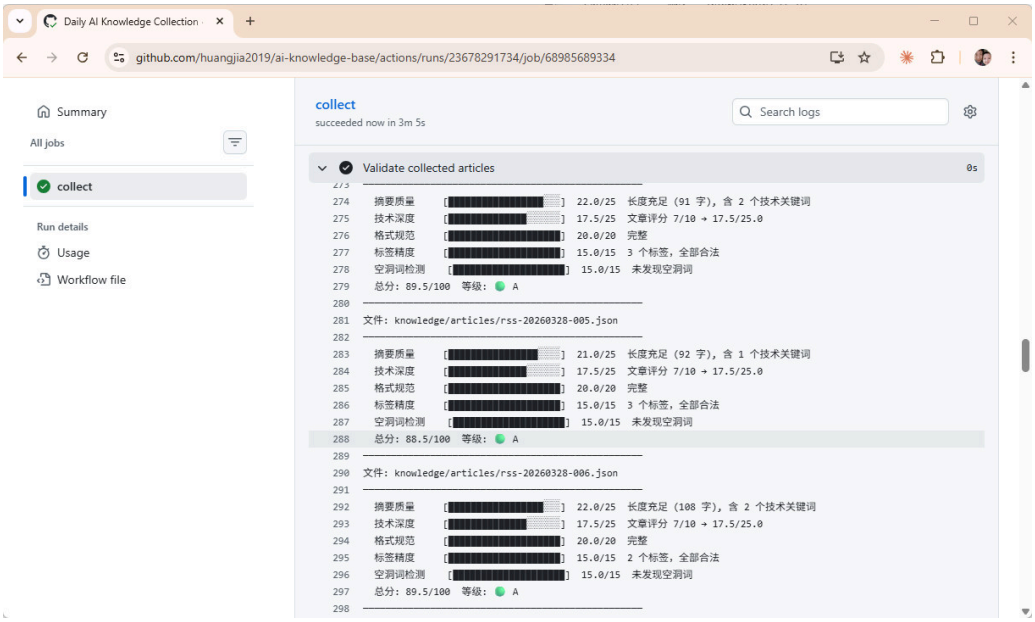
> 在 GitHub 网页上操作：点击 **Run workflow** 按钮，等待运行完成后会看到绿色 ✓ 标记。回到仓库首页可以看到 `github-actions[bot]` 自动提交的 commit。

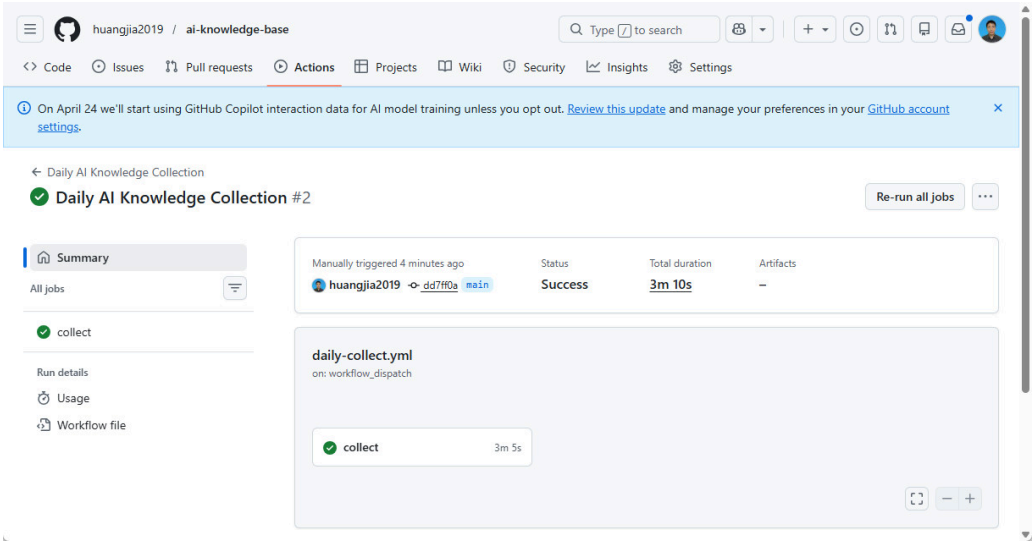
5. 回到仓库首页，查看是否有新 commit


Daily AI Knowledge Collection
github.com/huangjia2019/ai-knowledge-base/actions/workflows/daily-collect.yml
huangjia2019 / ai-knowledge-base
Type / to search
2
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
On April 24 we'll start using GitHub Copilot interaction data for AI model training unless you opt out. Review this update and manage your preferences in your GitHub account settings.
Actions
New workflow
All workflows
Daily AI Knowledge Collection
Management
Caches
Attestations
Runners
Usage metrics
Performance metrics
3
Daily AI Knowledge Collection
daily-collect.yml
Filter workflow runs
0 workflow runs
Event
Status
Branch
Actor
This workflow has a workflow_dispatch event trigger.
Run workflow
4
This workflow has no runs yet.


Daily AI Knowledge Collection
github.com/huangjia2019/ai-knowledge-base/actions/runs/23678003973/job/68985390703#logs
Daily AI Knowledge Collection #1
Cancel workflow
Latest #2
Summary
All jobs
collect
Run details
Usage
Workflow file
collect
Started 2m 12s ago
Search logs
Set up job 1s
Checkout repository 1s
Setup Python 2s
Install dependencies 2s
42 Successfully installed anyio-4.13.0 certifi-2026.2.25 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.11 python-dotenv-1.2.2 pyyaml-6.0.3 typing_extensions-4.15.0
Run collection pipeline 2m 6s
Validate collected articles
Commit and push results
Post Setup Python
Post Checkout repository


Daily AI Knowledge Collection
github.com/huangjia2019/ai-knowledge-base/actions/runs/23678003973/job/68985390703#logs
Summary
All jobs
collect
Run details
Usage
Workflow file
collect
failed now in 2m 56s
Search logs
Run collection pipeline 2m 48s
734 ##
Validate collected articles 0s
1 ▶ Run if ls knowledge/articles/*.json 1>/dev/null 2>&1; then
16 ==========
17 JSON 格式校验结果
18 ==========
19 [FAIL] knowledge/articles/test-bad.json
20 - 缺少必填字段: source_url
21 - 缺少必填字段: summary
22 - 缺少必填字段: tags
23 总计: 43 文件, 42 通过, 1 失败
24 Error: Process completed with exit code 1.
之前的测试文件导致一些校验失败
但我修改了Yml让流程继续执行。
Commit and push results 0s
Post Setup Python 0s
Post Checkout repository 1s
1 Post job cleanup.
2 /usr/bin/git version
3 git version 2.53.0
4 Temporarily overriding HOME='/home/runner/work/_temp/56e246a1-7a0c-4408-8881-920a720a9d8d' before making global

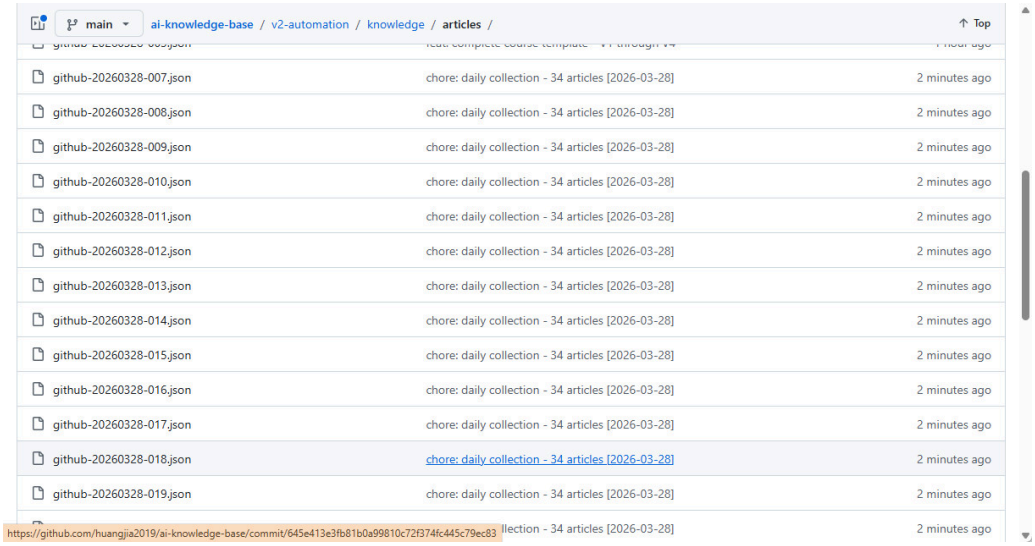




1.7 验证自动采集数据

```
# 拉取远程新数据
git pull

# 查看新采集的数据
ls knowledge/articles/
```



检查清单：

检查项	期望	实际
workflow 文件已推送	是	
Actions 页面能看到 workflow	是	
手动触发成功（绿色标记）	是	
Validate 步骤正常运行	是	
仓库有 bot 的新 commit	是	
commit 消息包含文章数量	是	
knowledge/articles/ 有新数据	是	

完成！ 你的知识库现在每天自动采集数据了。